Bias Detection in Shortlisting-Style Employment Systems: A Fairness Audit Using Benchmark and HR Data

A fairness audit using the Adult benchmark and a merged organizational HR dataset

Candidate: Parabhjyot Singh

Student number: GH1031307

Programme : Master's Computer Science

University: Gisma University Potsdam

Supervisor: Dr. Reza Babaei

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`GitHub repository` - "https://github.com/SeverusSnapee/M599_THESIS"

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Abstract

Automated screening systems can reduce recruitment workload and standardise early-stage decisions, but they can also reproduce historical inequality when trained on biased outcomes or when proxy variables encode protected characteristics. This dissertation evaluates group fairness in two employment-adjacent classification settings aligned to the implemented workflow: the

UCI Adult dataset and a merged organisational HR dataset assembled from three internal spreadsheets. Three baseline classifiers were trained using scikit-learn: logistic regression, random forest, and gradient boosting. Each model was evaluated using predictive accuracy alongside three fairness metrics: Statistical Parity Difference (SPD), Disparate Impact (DI), and Equal Opportunity Difference (EOD).

The study adopts a quantitative experimental design with transparent preprocessing, categorical encoding, stratified train-test splitting, and explicit post hoc fairness calculation. Rather than treating fairness as a single mathematical property, the dissertation approaches it as a multi-dimensional and context-dependent problem. The findings are therefore interpreted in relation to both the technical behavior of the models and the broader literature on algorithmic hiring, proxy discrimination, and sociotechnical systems. Bias mitigation techniques such as reweighing and equalised odds are reviewed as realistic future extensions rather than described as if they had already been implemented.

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1. Introduction

1.1 Background and context

Machine learning now plays a visible role in many decision-making systems that affect access to jobs, credit, housing, and public services. In recruitment and human resource management, algorithmic tools are often introduced on the promise that they can process information faster, standardise evaluation, and reduce the burden placed on human recruiters. In practice, however, the use of automated systems in employment settings raises a more difficult question: can a model be operationally efficient while still treating groups fairly? (Raghavan et al., 2020).

That question matters because hiring and employment decisions are not merely technical classifications. They affect income, career progression, stability, and access to social opportunities. When a decision pipeline is partially or fully automated, the potential scale of harm also changes. A biased human reviewer may affect a limited number of candidates; a biased automated filter can apply the same pattern to thousands. The point is not that human decision-making is neutral while algorithmic decision-making is biased. It is that algorithmic systems can make unequal treatment harder to notice while giving it the appearance of objectivity.

This concern has been central to the rise of fairness in machine learning as a research field. The core insight running through that literature is that machine learning systems learn from existing data rather than from neutral truths. If those data reflect past inequalities, the resulting model may reproduce them. This is especially relevant in employment contexts, where labels such as “successful employee,” “retained worker,” or “high performer” are not simple reflections of underlying merit. They are shaped by organizational practice, opportunity structures, and managerial judgement (Mehrabi et al., 2021; Suresh and Guttag, 2021).

Much of the public debate around automated hiring has focused on text-based systems that process resumes or application materials using natural language processing. That discussion is important, but fairness concerns are not limited to text. Structured variables such as education, occupation, salary, tenure, department, and location can also encode group-level inequality. For that reason, a fairness audit can be carried out meaningfully even in structured data settings, provided the scope of the claim is clearly stated. This dissertation takes that approach. It treats automated resume shortlisting as the motivating context, but it operationalises the problem through two structured datasets: the adult benchmark dataset and a merged organisational HR dataset (Becker and Kohavi, 1996; Raghavan et al., 2020).

1.2 Problem statement

A model can perform well in aggregate while still producing systematically uneven outcomes across demographic groups. This is the central problem addressed in the dissertation. In practical terms, a classifier may achieve high overall accuracy but still give one group fewer positive outcomes, or identify positive cases more reliably for one group than for another. When such a system is placed in an employment context, those disparities become ethically significant because they relate to access to work and related opportunities.

The problem is not simply one of data imbalance. It is also one of definition. Fairness in machine learning does not have a single universally accepted formal meaning. Different fairness metrics capture different harms. Some focus on equal selection rates, while others focus on equal error rates. A system that looks fair under one definition may fail under another. As a result, fairness cannot be treated as a box to tick with one number. It must be approached as a set of competing and context-sensitive judgements.

In employment settings, the problem is made more difficult by the nature of the target label. Labels are often proxies. Income, attrition, promotion, productivity, and managerial evaluation may all be shaped by historical and organisational inequalities. Therefore, even if a protected attribute is excluded from the model, the target and remaining features may still reproduce bias. This means that an adequate audit must examine not only inputs, but also outputs and the meaning of the labels being predicted.

1.3 Research aim

The aim of this dissertation is to evaluate whether standard baseline machine learning classifiers exhibit measurable group-level disparities when applied to structured employment-adjacent datasets. More specifically, the study seeks to develop and document a reproducible fairness-audit workflow using one benchmark dataset and one organisational dataset, and to interpret the resulting model behaviour through the lenses of fairness theory and employment-related machine learning.

A secondary aim is to maintain close alignment between the written dissertation and the code that was actually run. That matters because fair machine learning projects often suffer from a mismatch between conceptual discussion and implementation. In this dissertation, the description of preprocessing, modelling, and fairness evaluation is grounded in the executed Python workflow rather than in a hypothetical or inflated version of the project.

1.4 Research question

The central research question is: how do standard machine learning classifiers—logistic regression, random forest, and gradient boosting—balance predictive performance and group fairness across a benchmark dataset and a merged organisational HR dataset?

This overarching question can be broken into several related questions. How do fairness outcomes differ across model types? To what extent do performance gains coincide with wider or narrower fairness disparities? How does dataset context influence fairness outcomes? And what do these patterns suggest about the usefulness and limits of fairness metrics in employment-related machine learning?

1.5 Objectives

The first objective is to document a transparent and reproducible modelling pipeline for fairness auditing. The second is to compute predictive and fairness metrics for multiple baseline models. The third is to interpret those metrics critically rather than treating them as self-explanatory. The fourth is to compare the findings with the broader literature on algorithmic bias, hiring systems, and sociotechnical fairness. The fifth is to identify realistic directions for future mitigation and governance.

1.6 Contribution and scope

A further point worth stressing is that the value of a fairness audit lies partly in what it prevents. In many organisational settings, models are developed quickly under pressure to improve efficiency, and ethical review enters only after a system has already been normalised. A dissertation like this one serves a different purpose. It insists that fairness should be considered early, that assumptions should be documented, and that even modest baseline models deserve scrutiny before they are treated as operationally useful. That point is particularly relevant in recruitment, where decisions made at the screening stage may shape the composition of later candidate pools in ways that are difficult to reverse. An unfair early filter does not merely produce an unfair shortlist; it can also distort the broader decision process by making some candidates invisible from the outset.

For the same reason, scope discipline matters. It is tempting to overstate what structured benchmark and organisational datasets can tell us about automated resume shortlisting in the wild. This dissertation avoids that overstatement. Instead, it treats fairness auditing as a transferable methodological practice. The claim is not that the Adult and HR datasets reproduce all the complexity of industrial recruitment, but that they make it possible to examine how standard classifiers behave when group fairness is evaluated seriously rather than assumed. That is a smaller claim, but it is also a more defensible one, and it reflects the level of care expected in master's-level research.

The dissertation contributes in three main ways. First, it provides an implementation-matched fairness audit rather than a purely theoretical account. Second, it combines a recognised benchmark dataset with a smaller organisational dataset, thereby balancing methodological comparability with practical relevance. Third, it treats fairness as a multi-dimensional concept that must be interpreted in context rather than reduced to a single statistic.

The dissertation does not claim to model a full industrial resume-screening product. It also does not claim that fairness can be resolved entirely through metrics. Its contribution is more modest but still meaningful: it shows how fairness can be audited in a structured, reproducible way using standard classifiers and employment-adjacent data.

This dissertation also makes a practical contribution by showing that fairness auditing does not need to begin with highly complex models or large-scale production systems. In many real settings, organisations start with ordinary tabular data, limited documentation, and simple baseline models.

That makes it important to show how fairness can be examined in a realistic workflow rather than only in ideal research conditions. In that sense, the value of this study lies not only in the final metric values, but also in the way the audit process is structured and documented. By keeping the written explanation closely aligned with the executed implementation, the dissertation provides a clearer example of how fairness checks can be integrated into an applied machine learning project. This is especially relevant in employment-related contexts, where even small design choices in preprocessing, target definition, and evaluation can influence the interpretation of fairness. The study therefore contributes not just a set of results, but a practical and reproducible approach that can be adapted in similar structured-data settings.

2. Literature Review

2.1 Framing the field

Fairness in machine learning has become one of the defining topics in contemporary AI research because algorithmic systems increasingly shape decisions in areas where unequal treatment has serious consequences. Employment is a particularly sensitive domain because the allocation of work and the evaluation of workers influence income, status, and long-term opportunity. The literature does not present fairness as a single settled concept. Instead, it reveals an active field marked by competing definitions, different assumptions about harm, and contrasting views on what a technical intervention can realistically achieve.

One useful way to begin is to distinguish between a technical view of fairness and a sociotechnical view. In the technical tradition, fairness is usually formalized as a property of predictions, scores, or classification of errors. The goal is to measure and, if possible, constrain disparities between groups. In the sociotechnical tradition, fairness is treated as inseparable from the institutions, histories, and social relations within which a model is used. These perspectives do not simply oppose one another; they often address different levels of the problem. This dissertation draws on both. It uses formal group fairness metrics because they are useful for auditing outcomes, but it does not treat those metrics as complete descriptions of justice.

2.2 Sources of algorithmic bias

A central contribution of the fairness literature is the argument that bias can emerge at many stages of the machine learning lifecycle. Mehrabi et al. (2021) note that bias can arise in problem definition, data collection, feature selection, target construction, evaluation, and deployment. This matters because it challenges the simplistic view that bias is something that can be "removed" only by cleaning a dataset or omitting a protected attribute.

Historical bias is often the starting point. Datasets used in employment settings are shaped by prior social and organisational decisions. If previous hiring, pay, or promotion patterns were unequal, those inequalities may be reflected in the data and carried into the model. Historical bias is particularly difficult because it does not disappear simply by improving coding technique. A model may learn from discriminatory histories even when its developers act in good faith.

Representation bias occurs when some groups are missing, underrepresented, or represented in distorted ways. In workforce data, this may happen when there are relatively few women in technical leadership roles or relatively few minority employees in senior positions. A model trained on such data may struggle to represent those groups accurately and may generalise poorly to them. Representation bias is therefore not only a statistical issue but also an issue of whose experience becomes legible to the model.

Measurement bias arises when variables do not cleanly capture the concepts they are intended to measure. In employment-related datasets, many variables are imperfect proxies for deeper constructs. Educational attainment may reflect class advantage as much as skill. Salary history may reproduce earlier discrimination. Performance ratings may be influenced by evaluator bias or organisational politics. This matters because a model can appear technically sound while learning from variables that already contain inequitable social meaning.

Aggregation bias appears when a single model is trained across heterogeneous groups whose relationships between features and outcomes differ. In some cases, a model that works reasonably well on average may systematically underfit a minority group because its patterns are dominated by the majority group. Evaluation bias arises when the model is assessed using aggregate metrics that hide subgroup failures. Deployment bias enters when a model is used in a setting that differs from the one in which it was designed or when users interpret outputs in ways that exaggerate confidence.

Suresh and Guttag (2021) are especially valuable here because they frame harms across the entire machine learning lifecycle rather than confining them to final predictions. That broader perspective is important for this dissertation because it reinforces the idea that fairness auditing should not begin and end with metric calculation. Metrics reveal disparities, but they do not explain by themselves where those disparities come from or how serious they are in social terms.

2.3 Bias in hiring systems

Algorithmic hiring has become a focal point in fairness debates because employment decisions combine high stakes with a long history of unequal treatment. Hiring systems are often marketed as tools for reducing subjectivity, but the literature suggests that this promise is frequently overstated. Raghavan et al. (2020) argue that hiring technologies are shaped by decisions about what is predicted, how validity is established, and how vendor claims are presented. In other

words, fairness problems are often embedded in the design of the system long before a classifier is trained.

A central issue is target construction. In recruitment, labels such as “best candidate,” “successful hire,” or “future high performer” are not directly observed. They must be approximated through proxies such as retention, productivity, supervisor ratings, or promotion outcomes. Yet each of these proxies may already reflect unequal conditions. A model may therefore look fair with respect to its chosen label while still reinforcing unfairness at a broader level. This is one reason why fairness results must be interpreted relative to the target being predicted.

Proxy discrimination is especially important in hiring systems. Excluding protected attributes does not guarantee fairness if correlated features remain. Career gaps, school prestige, job history, postcode, or language use may all correlate with gender, race, disability, or class. Barocas and Selbst (2016) emphasise that anti-discrimination problems do not disappear when sensitive features are omitted, because other variables may still carry similar information. In practical terms, this means that a model can discriminate even when it never sees gender or race explicitly.

The public controversy around recruitment systems that allegedly disadvantaged women has made these concerns visible beyond academia. The often-cited Reuters report on Amazon’s experimental hiring tool became influential not only because of the specific example, but because it captured a broader social anxiety: the fear that AI systems may replicate inequality while being perceived as neutral. Although media examples are not substitutes for systematic research, they help explain why fairness in hiring remains such a politically and ethically charged topic.

Scholars working from a sociotechnical perspective have gone further, arguing that algorithmic hiring cannot be understood through narrow model evaluation alone. Drage and Mackereth (2022), for example, question whether AI genuinely “debiases” recruitment or simply reframes deeply political questions as technical optimisation problems. This critique matters because it warns against overclaiming. A fairness audit can reveal disparities in a classifier, but it cannot by itself settle whether the broader hiring process is just.

2.4 Fairness metrics and their tensions

2.4.1 Comparing fairness definitions in practice

A practical problem in fair machine learning is that fairness definitions are often discussed as though they were interchangeable when in fact they reflect different moral priorities. Statistical parity focuses on the distribution of favourable predictions. Equal opportunity focuses on the treatment of truly positive cases. Disparate impact expresses allocative disparity through a ratio rather than an absolute gap. In an employment context, those are not merely technical distinctions. They correspond to different questions about harm. A system that selects one group far less often than another may be troubling even if its true positive rates are relatively similar. Conversely, a model might show moderate overall selection parity while still recognising qualified candidates from one group less consistently than those from another.

The value of multi-metric evaluation is therefore not that it produces a more complicated table, but that it prevents the false impression that fairness can be decided by a single statistic. This is especially important in dissertations and applied reports, where a single fairness number can easily be mistaken for a final ethical judgement. In reality, metric choice is already a normative choice. It determines which kind of disparity becomes most visible. For that reason, the present dissertation treats SPD, DI, and EOD as complementary rather than redundant. The point is not to identify one universally correct metric, but to use several in order to show where models may look acceptable under one lens and questionable under another.

A further issue concerns operational thresholds. Many classification systems do not work directly with ground-truth categories alone, but with predicted probabilities that are converted into binary outcomes through some threshold. Yet fairness values can change meaningfully when that threshold changes. This matters because employment systems often manage workload through thresholding: stricter thresholds reduce the number of candidates flagged or shortlisted, while lower thresholds increase throughput. A fairness audit based on binary outputs is therefore always an audit of a particular decision configuration rather than of an abstract model in the absence of operational rules. The present study accepts that limitation and interprets its metrics as properties of the executed pipeline rather than universal features of the classifier.

Table 2.1: Comparison of the fairness metrics used in this dissertation.

Metric	Primary question	Useful for	Main caution
SPD	Are positive outcomes distributed differently across groups?	Screening for unequal access to favourable decisions	Does not consider whether predictions are correct
DI	How large is the proportional gap in positive outcomes?	Adverse-impact style interpretation in employment settings	Direction depends on group coding and still ignores true labels
EOD	Are true positive cases recognised equally across groups?	Assessing opportunity-style fairness	Depends on the meaning and quality of the positive label

The fairness metrics used in this dissertation—Statistical Parity Difference, Disparate Impact, and Equal Opportunity Difference—belong to the family of group fairness measures. Their appeal lies in the fact that they make disparities observable and comparable. However, the literature also makes clear that they capture different harms and do not always move together.

Statistical parity focuses on the rate of positive outcomes across groups. In practical terms, it asks whether one group is selected or classified positively more often than another. This is a useful metric when the primary concern is unequal allocation of opportunities. In employment settings, that concern is highly relevant because selection rates have direct distributive implications. Yet statistical parity ignores the true label. A model can satisfy parity in outcomes while still making uneven errors across groups.

Disparate Impact is closely related but expresses the disparity as a ratio rather than a difference. It is especially resonant in employment law because adverse impact analysis has often been framed in proportional terms. The ratio format can be easier to interpret in practical settings, but it shares the conceptual limitation of statistical parity in that it is insensitive to the correctness of predictions.

Equal Opportunity Difference captures a different fairness idea. Rather than asking whether one group receives more positive outcomes overall, it asks whether truly positive cases are recognised equally across groups. This criterion, associated with Hardt, Price and Srebro (2016), often feels normatively attractive because it aligns with intuitions about fair opportunity. In a hiring-like setting, it corresponds to the idea that qualified candidates should not be systematically overlooked because of group membership. However, EOD also depends on the

quality of the label. If the ground truth is itself socially biased or imperfect, parity in true positive rates may still amount to parity with respect to a problematic target.

The literature repeatedly warns that these fairness criteria can conflict. Chouldechova (2017) and Kleinberg, Mullainathan and Raghavan (2016) show that when base rates differ across groups, it is generally impossible to satisfy several desirable fairness properties simultaneously. This means that fairness evaluation is never a purely technical exercise. Choosing a metric is also choosing a normative focus. Is the main concern unequal selection rates? Unequal recognition of positive cases? Equalised error burdens? Each answer points to a different fairness criterion.

This dissertation therefore adopts a multi-metric approach. It does not assume that one metric settles the matter. Instead, SPD, DI, and EOD are reported together so that different fairness harms can be assessed side by side. This is consistent with both the literature and the practical reality that employment systems are rarely judged against one fairness principle alone.

2.5 The impossibility problem and the meaning of trade-offs

The impossibility results in the fairness literature have major implications for applied work. They show that in many realistic cases, fairness is not something a model can optimise perfectly along all desirable dimensions. When base rates differ, improving one criterion may worsen another. In practice, this means that fairness–accuracy trade-offs and fairness–fairness trade-offs are unavoidable.

This matters because applied machine learning often frames model selection as if there were a single best solution. The fairness literature disrupts that assumption. A model with higher predictive accuracy may exhibit stronger disparities on one fairness metric. A model adjusted to reduce disparate impact may lose calibration or increase false positives. Therefore, fairness cannot be treated as an engineering constraint that can always be added without cost.

For this dissertation, the practical lesson is that results must be interpreted through a trade-off lens rather than through a search for one “best” model. The question is not simply which model is most accurate or which one is fairest according to a single measure. The more meaningful question is what kinds of harm each model appears to create or reduce, and how those harms compare with the practical and legal expectations of employment-related decision-making.

2.6 Bias mitigation methods

Bias mitigation methods are commonly grouped into pre-processing, in-processing, and post-processing categories. Pre-processing methods change the data before training. Their advantage is that they can be used with standard downstream classifiers. Reweighing, introduced by Kamiran and Calders (2012), is one of the best-known examples. It adjusts the effective importance of training instances to reduce dependency between protected groups and target outcomes. This approach is attractive because it is comparatively simple and model-agnostic.

In-processing methods alter the learning objective itself, for example by adding fairness constraints or penalties. These approaches can integrate fairness directly into model optimisation, but they are often more complex to implement and interpret. Post-processing methods change the final decision outputs after training. Equalised odds and equal opportunity post-processing methods fall into this category. Their main advantage is that they can be applied even when the underlying model is fixed. Their main limitation is that they may require explicit use of group membership at decision time, which raises legal and organisational questions.

The mitigation literature is valuable for this dissertation even though mitigation was not executed. It shows that fairness interventions are possible, but also that they involve choices rather than neutral corrections. Improving one metric may worsen another. Reducing disparity may lower conventional performance. This is why the present dissertation treats mitigation as future work rather than as if it were already implemented. The baseline must be clear before any claim about improvement can be made.

2.7 Toolkits and standardisation

2.7.1 Legal and regulatory context

The legal and regulatory context gives additional weight to fairness auditing in employment-related systems. Technical fairness metrics do not map perfectly onto anti-discrimination law, but they do intersect with concerns that the law has long recognised, particularly around practices that appear neutral while disadvantaging some groups disproportionately. This is one reason ratio-style reasoning remains important in applied fairness work. A metric such as disparate impact does not by itself establish legal liability, yet it can function as a practical early-warning signal that a model's outputs deserve closer scrutiny.

In the UK and European context, equality law and data protection law both shape how employment-related automation should be approached. The Equality Act 2010 is concerned with direct and indirect discrimination, while data protection frameworks raise questions about transparency, accountability, and the appropriate use of personal data in consequential automated systems. The GDPR and emerging AI governance frameworks do not tell a developer which fairness metric to adopt, but they do reinforce the expectation that organisations should be able to justify their data use, explain what their systems are designed to do, and demonstrate that risks have been examined rather than ignored (European Parliament, 2016; Information Commissioner's Office, 2020; European Commission, 2021).

This matters for the present dissertation because it shows that fairness metrics are not just academic abstractions. In employment settings, evidence of disparity may have reputational, governance, and compliance implications even before any formal complaint or legal challenge arises. A technically responsible organisation should therefore be able to explain what its model predicts, why that target was chosen, what forms of disparity were checked, and how those checks will be revisited as the system or the workforce changes over time. The wider lesson is that fairness auditing is valuable not only because it improves model evaluation, but also because it supports accountable organisational practice.

AIF360 and Fairlearn have become important because they standardise the calculation of fairness metrics and provide reference implementations of mitigation methods. Their practical value is significant: they reduce ambiguity, improve reproducibility, and make fair machine learning more accessible to practitioners. At the same time, the existence of such toolkits does not remove the need for critical judgement. A metric computed through a library call is not automatically the right metric for the application. Nor does a mitigation method guarantee justice simply because it is technically implemented (Bellamy et al., 2018; Bird et al., 2020).

This dissertation uses custom fairness calculations aligned to the executed code rather than using a toolkit directly. That choice improves transparency and keeps the analysis closely tied to the actual script. The literature and toolkit documentation remain relevant, however, because they provide the conceptual basis for the metrics and the future mitigation pathways discussed later.

2.8 Research gap and relevance to this dissertation

One area that deserves fuller attention is the legal and regulatory context within which fairness in employment systems is discussed. Although legal standards and technical fairness metrics do not

map perfectly onto one another, they intersect in important ways. Employment law has long been concerned not only with explicit differential treatment but also with practices that appear neutral while disadvantaging certain groups disproportionately. This is why ratio-based measures such as disparate impact remain so influential in applied fairness work. They do not determine legal liability on their own, but they provide a structured way of asking whether model outputs may have consequences that resemble established forms of indirect discrimination. In a dissertation concerned with automated shortlisting, this connection matters because it shows that fairness metrics are not merely abstract research tools; they also sit alongside real legal and compliance concerns.

The UK and European regulatory environment strengthens this point. Data protection law, equality law, and emerging AI governance frameworks all point toward the need for transparency, accountability, and justification when automated systems affect individuals significantly. Even where fully automated decision-making is not used, decision support systems can shape outcomes in ways that should be open to scrutiny. For employment-related models, this means organisations should be prepared to explain not only how a system performs, but also what it is trained to predict, what data it uses, what disparities have been checked, and how those checks will be revisited over time. The legal literature therefore reinforces a key theme of this dissertation: fairness auditing should be embedded in governance rather than treated as a one-off technical exercise.

Another debate in the literature concerns whether benchmark datasets still have value given their distance from live organisational settings. Some scholars criticise benchmark-heavy fairness research for becoming detached from real social processes. That criticism is partly justified. Benchmarks can flatten context and encourage the impression that fairness is a universal optimisation problem. Yet benchmark datasets also remain useful because they allow researchers to compare methods, reproduce results, and study disparities on shared ground. The Adult dataset continues to matter precisely because it occupies this benchmarking role. Its usefulness lies not in mirroring every aspect of real-world hiring, but in providing a stable point of comparison. The stronger position, therefore, is not to reject benchmark datasets outright, but to pair them with contextual reflection and, where possible, with more applied data sources. That is the approach taken here.

The literature also raises an important practical question about what counts as evidence of improvement. In fair machine learning, it is tempting to focus on whether a mitigation technique shifts a fairness metric in the desired direction. Yet a narrow focus on metric movement can be misleading. If a fairness intervention reduces SPD but obscures the interpretability of the model, introduces instability, or changes error burdens in ways that are harder to justify, then its value is not straightforward. This is why the literature increasingly stresses documentation, transparency, and human review alongside numerical fairness evaluation. A model is not made trustworthy by metric movement alone. Trustworthiness depends on whether those metric changes can be explained, defended, and governed in practice.

Another important strand of the literature concerns the relationship between fairness and explanation. Some scholars assume that more interpretable systems are therefore easier to govern fairly. There is some truth in this: interpretable models may make it easier to identify problematic variables, justify decisions, and challenge inappropriate outputs. Yet the literature also shows that interpretability is not equivalent to fairness. A transparent model can still be discriminatory, while a less transparent model might in some circumstances distribute outcomes more evenly. What matters is not whether a model can be explained in general terms, but whether the explanation is relevant to the harm being investigated. In employment contexts, the demand for explanation is especially strong because affected individuals may reasonably expect to know why they were filtered out or ranked lower. This suggests that fairness research in hiring should not be detached from questions of explanation, contestability, and procedural justice.

A related debate concerns whether fairness should be framed as a property of predictions, scores, or decision procedures. Some metrics, including the ones used in this dissertation, are usually applied to final binary predictions. Yet many real systems operate through scores, thresholds, ranking stages, and human review layers rather than single one-shot classifications. This raises a methodological issue for the literature: fairness measured at one stage of the pipeline may not capture the full pattern of disparity across the workflow. For example, a model might produce relatively even binary outcomes at one threshold while still assigning systematically lower scores to one group, thereby affecting downstream ranking or review intensity. Although the present dissertation does not model score-level or threshold-level fairness in detail, this literature is important because it highlights the limits of purely end-point evaluation. It also suggests a clear

direction for future work: extending fairness audits from final class labels to ranked decision processes and score distributions.

The literature on labour markets also helps explain why fairness in employment systems cannot be interpreted in isolation from broader social structure. Occupational segregation, unequal care burdens, educational stratification, and uneven access to professional networks all shape workforce outcomes long before data enter a machine learning pipeline. This means that even a technically well-measured fairness disparity may point to a deeper social pattern that the model is merely reflecting rather than creating from nothing. That observation should not be used to excuse model bias. On the contrary, it is one reason such models require careful scrutiny. If a classifier simply formalises existing disadvantage and scales it efficiently, then its technical sophistication becomes part of the problem rather than part of the solution.

The legal literature adds another layer of complexity. Anti-discrimination law has historically focused on both direct discrimination and indirect discrimination. In machine learning terms, direct discrimination roughly corresponds to explicitly treating protected groups differently, while indirect discrimination corresponds more closely to the use of apparently neutral practices that disadvantage certain groups disproportionately. This legal distinction maps imperfectly onto technical fairness metrics, but the overlap is instructive. Metrics such as disparate impact are useful partly because they offer a bridge between computational evaluation and established concerns in employment law. At the same time, the law does not provide a simple metric-based answer. Legal assessment depends on context, justification, proportionality, and evidentiary standards. Therefore, the literature suggests that fairness metrics are best understood as part of a broader evidentiary picture rather than as self-sufficient proof of unlawful discrimination.

There is also an important methodological gap between benchmark-centred fairness studies and organisational case studies. Benchmark datasets are valuable because they make comparisons possible, but they often flatten the social meaning of the task. Organisational datasets bring the analysis closer to practical decision environments, yet they are often smaller, messier, and harder to interpret. This tension is one of the main reasons the present dissertation uses both. The combination is methodologically useful because it allows the study to move between comparability and realism without pretending that either dataset type is sufficient on its own.

Three gaps in the literature are especially relevant here. The first is the gap between fairness theory and implementation. Many applied projects discuss fairness at length but do not align their narrative clearly with what was actually executed. The second is the gap between benchmark fairness studies and real organisational datasets. Benchmark datasets such as Adult are useful for comparison, but they do not capture the full messiness of workplace data. Organisational datasets are more realistic but often limited in scale and interpretability. The third is the gap in explicit discussion of target meaning, especially when labels are engineered from organisational records.

This dissertation addresses those gaps by keeping implementation and narrative aligned, by combining a benchmark dataset with a smaller organisational dataset, and by being explicit about what the HR target represents and what it does not represent. The result is a study that is modest in scope but careful in method. Rather than claiming to solve fairness in hiring, it demonstrates how a structured fairness audit can be carried out in a way that is technically transparent and conceptually honest.

3. Methodology

3.1 Methodological approach

This dissertation uses a quantitative experimental design centred on supervised binary classification. That design is appropriate because the project seeks to compare how several baseline machine learning models behave in relation to both predictive performance and group fairness. A quantitative design allows the variables of interest—model type, dataset, accuracy, and fairness metrics—to be defined explicitly and compared systematically. It also supports reproducibility, which is especially important in fairness research where preprocessing choices, train-test splits, and group definitions can materially affect outcomes.

At the same time, the study does not treat fairness as a purely numerical problem. The quantitative core of the methodology is supplemented by critical interpretation throughout the dissertation. This is important because fairness metrics are not self-explanatory and because the meaning of the results depends on the target label, the dataset context, and the broader employment setting in which such systems might be used.

A baseline-first design was chosen deliberately. Before discussing mitigation, it is necessary to document how ordinary models behave on the available data. This baseline makes later fairness claims more credible because it shows what the unadjusted system does before any corrective method is introduced. For a master's dissertation, this also keeps the study methodologically coherent and realistically executable.

3.2 Dataset choice and study design

The dissertation uses two datasets: the UCI Adult dataset and a merged organisational HR dataset. This dual-dataset design serves two purposes. First, it allows the study to include a recognised fairness benchmark, which helps situate the work within an established literature. Second, it allows the study to test the same audit workflow on a smaller organisational dataset, bringing the analysis closer to practical employment data (Becker and Kohavi, 1996).

The Adult dataset is widely used in fairness research because it contains clear demographic attributes and a well-known binary target: whether annual income exceeds fifty thousand US dollars. It is not a hiring dataset, but it is useful as a benchmark because it contains employment-related and demographic variables often implicated in fairness studies (Becker and Kohavi, 1996; Fairlearn documentation, n.d.).

The use of these datasets is sufficient for the aims of the study, but it also places limits on generalisation. The dissertation does not claim that the findings transfer directly to all real-world shortlisting systems. Rather, it argues that the datasets are adequate for evaluating a fairness-audit pipeline in structured employment-adjacent classification tasks.

3.3 Data preparation and preprocessing

3.3.1 Why preprocessing choices matter for fairness

Preprocessing is often treated as background cleaning, but in fairness research it should be regarded as part of the substantive analysis. Decisions about which observations are removed, how categories are encoded, what counts as missing, and how the target is defined can all affect

the fairness profile of the final model. This dissertation therefore treats preprocessing not as a neutral technical stage but as one of the points where fairness assumptions become operational.

The Adult dataset illustrates this clearly. Removing rows that contain ambiguous values simplifies the workflow and is common in benchmark studies, but it may also change subgroup composition if missingness is patterned rather than random. The HR dataset poses a different challenge. Dropping identifiers and target-revealing fields is necessary to prevent leakage, but the removal of date fields also means that variables such as tenure or age are not engineered into the final model, even though they may be substantively relevant. These trade-offs do not make the implementation wrong; they simply show why preprocessing has to be documented as part of the fairness story rather than hidden behind a few lines of code.

The same applies to the choice of label encoding. For tree-based models, label encoding is often acceptable in practical terms, but for logistic regression it may create an artificial ordinal structure among nominal categories. This is one reason the dissertation repeatedly emphasises implementation fidelity. A more theoretically refined preprocessing strategy might have been possible, but describing a different pipeline from the one that was actually executed would have reduced the integrity of the study. In this project, transparent alignment was the more important methodological priority.

Table 3.2: Key preprocessing decisions and their implications for fairness interpretation.

Decision point	**Implemented choice**	**Reason for choice**	**Fairness implication**
Adult missing values	Replace '?' and drop incomplete rows	Keeps pipeline simple and reproducible	May alter subgroup composition if missingness is patterned
HR target	Engineer target from termination_date	Observable and reproducible binary outcome	Fairness is measured relative to attrition, not merit
Categorical encoding	LabelEncoder on object columns	Matches executed workflow and keeps code compact	May impose ordinal structure for linear models
Leakage control	Drop identifiers, names, termination-related fields,	Prevents unrealistic prediction shortcuts	Improves validity of fairness comparison

	raw dates		across models

Preprocessing was a major methodological stage because the fairness profile of a model is influenced not only by model choice but by how the data are prepared. In the Adult dataset, missing values represented by question marks were treated as missing and removed. This is a common choice in benchmark studies because it simplifies categorical processing and avoids adding uncertain assumptions through imputation.

For the HR dataset, preprocessing involved both merging and cleaning. HRDatabase.xlsx, EmployeeInformation.xlsx, and DepartmentInformation.xlsx were merged using id and department keys. Once combined, the dataset contained both demographic and organisational features such as gender, education, position, department, performance review, city, salary, overdue vacation status, and manager. The target was then engineered from the termination_date field by labelling active employees differently from terminated employees.

A crucial part of preprocessing involved removing leakage-prone variables. Fields such as employee identifiers, names, termination_date, termination_reason, and raw date fields were removed before modelling. This was necessary because such variables either contain direct information about the target or are unsuitable for direct inclusion in the selected classifiers. The removal of these fields improved methodological validity by ensuring that the models relied on plausible predictive features rather than outcome-revealing artefacts.

The dissertation also standardised column names and ensured a fully numeric modelling table prior to training. These choices might appear technical, but they matter because machine learning workflows are highly sensitive to unclean schemas and inconsistent feature representations.

3.4 Encoding strategy

All object-type categorical variables were encoded numerically using LabelEncoder. This reflects the actual implementation and therefore preserves alignment between the dissertation and the executed code. The advantage of this approach is simplicity: it transforms all categorical features into machine-readable form without requiring a more elaborate preprocessing pipeline.

The use of label encoding, however, deserves critical comment. For tree-based models, such encoding is often workable because the model learns splits rather than linear weights. For logistic regression, the situation is less ideal because integer labels can create an artificial ordinal

interpretation among nominal categories. A more theoretically refined study might therefore prefer one-hot encoding for nominal variables. Even so, retaining label encoding was the correct choice for this dissertation because the methodological priority was fidelity to the implemented workflow rather than retrospective optimisation.

This is an example of a broader principle: in a dissertation of this kind, transparency and alignment are more valuable than presenting an idealised pipeline that was never actually run.

3.5 Train-test split and class balance

Both datasets were split into training and testing sets using scikit-learn's train_test_split with stratification. Stratification was important because each dataset had an imbalanced target distribution. In the Adult dataset, the high-income class is smaller than the low-income class. In the HR dataset, the terminated class is smaller than the active class. Preserving class proportions across train and test sets improves the stability of both accuracy and fairness estimates.

A fixed random state was used for reproducibility. The study relies on one executed split rather than repeated cross-validation. This is a limitation, but it is also consistent with the dissertation's emphasis on implementation alignment. Repeated splits or threshold sweeps would be valuable future extensions, especially for the smaller HR dataset, but they were not part of the executed workflow and are therefore not retroactively inserted into the methodology narrative.

3.6 Model selection

The three selected models—logistic regression, random forest, and gradient boosting—provide a useful range of baseline behaviour. Logistic regression acts as the simplest and most interpretable baseline. Its linear form makes it a useful starting point because it reveals whether fairness concerns are already visible without relying on highly flexible modelling. Random forest introduces non-linear structure and interaction modelling while remaining a common and robust tabular classifier. Gradient boosting adds a stronger ensemble approach capable of capturing more subtle patterns.

These models were chosen for practical as well as theoretical reasons. All three are widely used, supported in scikit-learn, and appropriate for tabular datasets. Together, they allow the dissertation to examine whether changes in model expressiveness are associated with changes in fairness outcomes.

3.7 Fairness metrics and their role in the design

The dissertation reports three fairness metrics: Statistical Parity Difference, Disparate Impact, and Equal Opportunity Difference. They were selected because they capture different dimensions of group-level disparity. SPD measures differences in the rate of positive predictions between groups. DI expresses a similar disparity in ratio form. EOD measures the difference in true positive rates between groups and therefore speaks to equal opportunity in recognising positive cases (Hardt, Price and Srebro, 2016).

These metrics were computed post hoc using a custom function. This function groups predictions by the protected attribute and derives each measure directly from observed outcomes. One important implementation detail is that the interpretation of DI depends on how the protected groups are encoded. Because the custom code computes the prediction rate of group 1 relative to group 0, and because encoding typically maps female to 0 and male to 1, DI values above one indicate a higher positive rate for the privileged group. This is not a mathematical problem, but it makes explicit documentation essential.

The dissertation reports fairness metrics alongside accuracy because fairness without performance context can be misleading. A model that appears fair but performs badly is not automatically acceptable, just as a highly accurate model is not acceptable if it imposes serious group disparities.

3.8 Ethical and governance considerations

The ethical and governance dimension of the methodology also has a practical side. In many organisations, the technical team that builds a model is not the same group that signs off on its operational use. This split creates a risk that fairness is treated as someone else's problem—either as an afterthought for compliance teams or as a vague ethical concern outside the remit of data scientists. The methodology adopted here implicitly rejects that separation. By documenting target construction, coding decisions, dataset limitations, and fairness calculations in one coherent workflow, the dissertation models an integrated approach in which technical and governance questions are considered together rather than sequentially.

A second point concerns the relationship between reproducibility and accountability. Reproducibility is often discussed as a scientific virtue, but in fairness work it also has an ethical dimension. A fairness claim that cannot be retraced or checked is difficult to challenge

responsibly. If an organisation were to say that its hiring model is fair without being able to show how the claim was reached, the assertion would have limited practical value. For that reason, the methodological choices in this dissertation—fixed random state, explicit preprocessing, saved artefacts, documented metric direction—support not only replication but accountability. They make it possible for another reader to inspect the chain of reasoning from data to result.

The organisational HR dataset contains potentially sensitive information. Even when handled for academic purposes in aggregate form, such data require caution. For that reason, the study treats the HR data as pseudonymised and does not report any individual-level conclusions. The analysis focuses on model behaviour at a grouped level rather than making claims about particular employees.

There is also an ethical issue in target construction. Predicting retention or termination does not amount to predicting merit. This means that the fairness audit is tied to the chosen label and cannot be interpreted as a broader moral judgement about who deserves opportunity. A responsible methodology has to make this explicit.

At a governance level, the study also distinguishes clearly between auditing and deployment. The models are not presented as tools ready for operational use in hiring or HR decisions. They are presented as objects of evaluation. This distinction matters because fairness auditing is meaningful partly as a way of preventing premature deployment.

3.9 Methodological limitations

The main limitations of the methodology are the use of structured datasets rather than direct shortlisting data, the small size of the HR dataset, the engineered nature of the HR target, the focus on a single protected attribute per dataset, and the absence of mitigation or repeated randomized evaluation. These limitations constrain what the dissertation can claim, but they do not undermine its core contribution. Instead, they define the project's scope clearly and honestly.

Another limitation is that fairness analysis is based on binary group comparisons, which simplifies a more complex social reality. In practice, disadvantage does not always operate along a single dimension, and people may be affected differently depending on the interaction between gender, race, age, class, disability, or migration background. This dissertation does not attempt an intersectional analysis because the available data and project scope did not support that level of complexity. However, this limitation is important because a model can appear relatively balanced under a single protected attribute while still producing unequal outcomes across more

complex group combinations. A related limitation is that the audit is carried out on final binary predictions rather than ranked outputs or probability scores. In a real shortlisting system, decisions may be influenced not only by whether a candidate is selected, but also by how candidates are ranked or prioritized for review. The present study therefore provides a useful baseline, but not a complete account of every fairness issue that may arise in operational recruitment systems.

3.10 Chapter summary

An additional methodological justification concerns the decision to focus on group fairness rather than individual fairness or causal fairness. Individual fairness, often summarised as the idea that similar individuals should be treated similarly, is attractive at the level of principle, but it requires a defensible account of what counts as similarity. In employment data, that

requirement is extremely difficult to satisfy because there is rarely a neutral or uncontested distance function over candidates or employees. Causal fairness approaches are also valuable, especially when researchers want to reason about counterfactual worlds in which sensitive attributes are changed while other variables are held constant. However, such approaches demand stronger structural assumptions and richer domain knowledge than were available for the present project. Group fairness was therefore selected not because it exhausts the fairness problem, but because it is the most appropriate approach for the available data, the project scope, and the need for transparent reproducibility.

The dissertation also deliberately avoids extensive hyperparameter optimisation. This may appear conservative, but it is methodologically defensible. Hyperparameter tuning could improve performance, yet it would also make the analysis more difficult to interpret because changes in fairness outcomes might then reflect tuning idiosyncrasies rather than broader differences between model classes. By keeping the models close to their baseline form, the study produces a cleaner comparison and reduces the risk of overfitting the dissertation to a particular set of performance targets. This is particularly important in the HR dataset, where the sample is small and highly tuned models would be more vulnerable to instability.

Another important justification concerns the use of a hold-out test set rather than embedding the fairness analysis entirely within cross-validation. A repeated cross-validation design would indeed offer more robust estimates, especially for a small dataset, but it would also generate a more complicated reporting problem because fairness metrics would need to be aggregated across folds and interpreted in relation to fold-specific subgroup distributions. For the purposes of a master's dissertation focused on implementation fidelity, the use of a single stratified train-test split is a reasonable compromise. It keeps the workflow understandable, reproducible, and closely tied to the executed code, while still allowing the study to reveal meaningful differences between models.

The methodology also treats documentation itself as part of rigor. In fair machine learning, undocumented assumptions are often where the most serious interpretive problems begin. This is why the dissertation repeatedly states what the target means, how the protected attribute is encoded, and what the fairness metrics do and do not imply. Such documentation is not merely

good style; it is an essential part of research validity in a domain where small implementation choices can change the meaning of the findings.

The methodology adopted in this dissertation is quantitative, comparative, and implementation aligned. It combines two structured datasets, three baseline classifiers, and three fairness metrics within a reproducible machine learning workflow. The chapter has shown why these choices are appropriate, what assumptions they involve, and where their limits lie. This provides the basis for the practical implementation described in the next chapter.

Table 3.1: Comparative overview of the datasets

Aspect	Adult dataset	Merged HR dataset	Why it matters
Primary role in study	Benchmark fairness dataset	Applied organisational case	Balances comparability and practical relevance
Rows used in modelling run	30,162	90	Shows strong benchmark scale versus small real-world sample
Target	Income above or below $50,000	Engineered from termination status	Highlights the importance of label meaning
Protected attribute audited	Sex	Gender	Supports dual-dataset group fairness comparison
Main limitation	Not a direct hiring dataset	Small and context-specific	Constrains generalisability of findings

Table 3.1: Comparative overview of the Adult and merged HR datasets used in the dissertation.

4. Implementation

4.1 Translating design into code

This chapter explains how the methodological design was translated into an executable workflow. The purpose here is not simply to reproduce code but to show how the conceptual decisions described earlier were operationalised in practice. In fair machine learning, implementation details matter because fairness outcomes can shift with changes in preprocessing, encoding, group assignment, or evaluation logic. For that reason, implementation

should be treated as a substantive part of the research rather than as an appendix to the “real” analysis.

The workflow used in this project can be divided into six stages: loading and inspecting the datasets, merging the HR files, engineering the HR target, cleaning and encoding features, splitting the data into training and test sets, and finally training and evaluating the models. This sequence reflects an ordinary tabular machine learning pipeline, but it is presented here with special attention to fairness-relevant choices.

4.2 Software environment

The implementation was carried out in Python using pandas, NumPy, scikit-learn, matplotlib, and openpyxl. pandas handled loading, merging, cleaning, and inspection of the datasets. NumPy supported numerical operations and missing-value handling. scikit-learn supplied the train-test split function, label encoder, classifiers, and basic evaluation tools. matplotlib was used for bar charts showing model accuracy and fairness metrics. openpyxl enabled the import of the Excel-based HR files.

This stack was chosen because it is widely used, well documented, and sufficient for reproducible experimentation on structured data. It also means the work can be rerun without specialised or proprietary tools.

4.3 Adult dataset implementation

The Adult dataset was loaded with explicit column names and then inspected by printing the first rows and the full list of columns. This initial inspection step is important because it verifies that the schema is being interpreted correctly before any modelling begins. Missing or ambiguous values denoted by question marks were replaced and removed. Column names were standardised by converting them to lower case and using underscores.

The target variable income was converted into a binary label. The protected attribute was defined as sex. After these steps, the dataset was encoded numerically and split into features and labels. The target distribution was printed before splitting so that class imbalance was visible.

4.4 HR dataset implementation

The HR data required additional preparation because it existed across three files. HRDatabase.xlsx, EmployeeInformation.xlsx, and DepartmentInformation.xlsx were loaded

separately, and their schemas were inspected before merging. The first merge joined the HR and employee information tables on the id field. The second merge joined department information using the department field. The resulting dataframe was then inspected again to confirm that columns and datatypes looked sensible.

The target variable was engineered from termination_date. If an employee had no recorded termination date, the target indicated active status; otherwise the target indicated termination. This produced a binary classification outcome that was practical, observable, and reproducible, even though it remained only a proxy for an employment-related target.

After target engineering, the implementation removed identifiers, names, target-revealing fields, and raw dates. Any remaining datetime columns were dropped. This step was essential for preventing leakage and creating a cleaner modelling table.

4.5 Encoding and modelling table

A helper function applied LabelEncoder to all object columns in both datasets. This made the modelling tables fully numeric. The encoded datatypes were then printed to verify the success of the transformation. While the use of label encoding is not ideal for every model, it reflects the actual implementation and keeps the dissertation closely tied to what was executed.

The final Adult modelling table included the standard benchmark features together with the binary target. The final HR modelling table included gender, education, position, department, performance_review, city, salary, overdue_vacation?, manager, and target.

4.6 Train-test splitting

Both datasets were split into training and testing sets using train_test_split with a fixed random_state and stratification by the target. This ensured reproducibility and helped preserve class balance across the splits. Given the small size of the HR dataset, stratification was especially important. Without it, there was a greater risk that the smaller class would be underrepresented in either the training or testing subset.

4.7 Model training

Three models were defined in a shared dictionary structure: logistic regression, random forest, and gradient boosting. Each was fitted on the training set and then used to predict the test set. Accuracy was calculated for each model. The use of a common training loop means that model

comparisons are based on a consistent evaluation structure rather than on separate ad hoc code paths.

The logistic regression model used an increased maximum iteration setting because the default iteration count was insufficient for convergence. The warning produced during fitting was not ignored in the dissertation because optimisation behaviour is relevant to how baseline models operate in practice.

4.8 Fairness calculation

Fairness metrics were calculated through a custom function that received the true labels, predicted labels, and the protected attribute values from the test set. This function grouped the data according to the encoded protected categories and then computed SPD, DI, and EOD from group-specific prediction rates.

The custom function improved transparency because every part of the calculation remained visible within the project. At the same time, it required explicit discussion of metric direction. Because the implementation compares group 1 to group 0, and because encoding usually maps male to 1 and female to 0, metric interpretation depends on that coding. Making this explicit helps avoid the common mistake of reporting fairness metrics as though they had a self-evident direction.

4.9 Results storage and plotting

After model evaluation, each set of outputs was appended to a results structure and converted into a dataframe. This dataframe included dataset, model, accuracy, SPD, DI, and EOD. It was then saved as baseline_results_with_fairness.csv. Saving the output in this way improves reproducibility and supports later reporting.

Bar charts were created for accuracy, SPD, DI, and EOD for each dataset. These plots were saved as PNG files with consistent names. This made it easy to insert them later into the Results chapter. The charts are intentionally simple because the goal is comparability and clarity rather than visual complexity.

4.10 Challenges encountered

The implementation involved several practical challenges. There were initial issues around Excel dependencies, datatype handling, and target construction in the HR workflow. The small size of

the HR dataset also made class balance a recurring concern. One of the useful features of the project was that these issues were not hidden. They were worked through openly, and the final dissertation remains aligned with the actual resolved pipeline rather than with an idealised description.

4.11 Reproducibility

One practical feature of the implementation worth emphasising is the repeated use of console inspection throughout the workflow. The script prints previews of the datasets, lists of columns, datatypes, and target distributions before and after key transformation steps. In ordinary programming terms this may look routine, but in dissertation work it is important because it makes the process auditable. It demonstrates that the researcher checked the assumptions built into the code rather than merely running a pipeline end to end and accepting the outputs at face value. In fairness research, that habit matters because a single mistaken assumption about group coding, missingness, or column selection can alter the interpretation of every metric that follows.

A second practical point concerns the relationship between implementation simplicity and analytical value. The workflow used here is not complicated by design. It does not rely on custom neural architectures, automated hyperparameter searches, or heavy pipeline abstraction. This should not be mistaken for a weakness. In fact, the relative simplicity of the implementation improves interpretability because each stage of the process remains visible. It also makes the fairness audit easier to explain and reproduce. In a dissertation context, this kind of disciplined simplicity can be a strength, particularly when the aim is to produce a trustworthy baseline rather than an over-engineered system.

The generation of saved plots also serves a methodological purpose beyond presentation. By exporting accuracy and fairness measures into stable visual artefacts, the implementation creates a direct bridge between the code and the final dissertation. This matters because it reduces the chance of transcription error or retrospective adjustment during write-up. The figures that appear in the finished thesis can be tied directly to the executed outputs, which strengthens the integrity of the reporting process.

A strength of the implementation is that it can be rerun from a single script with clearly defined inputs and outputs. Dependencies are limited and standard, intermediate checks are visible through console output, and results are exported into reusable artefacts. In fairness research,

reproducibility is especially valuable because the interpretation of disparities depends heavily on documented implementation decisions.

Table 4.1: Baseline model configuration and evaluation design

Model / element	Implementation	Purpose in study	Main caution
Logistic Regression	scikit-learn baseline with increased max_iter	Linear and interpretable benchmark	May underfit non-linear patterns; convergence sensitivity
Random Forest	scikit-learn tree ensemble	Captures non-linear structure and interactions	Can learn proxy relationships through complex splits
Gradient Boosting	scikit-learn GradientBoostingClassifier	Higher-capacity tabular baseline	May overfit small datasets or intensify disparity
Evaluation design	Stratified train-test split; accuracy + SPD/DI/EOD	Consistent comparison across models and datasets	Single executed split limits robustness

Table 4.1: Baseline model configuration and evaluation design.

5. Results

5.1 Reporting conventions

This chapter presents the executed outputs exactly as produced by the modelling script. Accuracy is reported as the baseline performance measure. Fairness is reported using Statistical Parity Difference (SPD), Disparate Impact (DI), and Equal Opportunity Difference (EOD). In this implementation, the protected group coding means that the sign and ratio direction of the fairness metrics must be read with care. In particular, because the custom metric function computes group 1 relative to group 0, DI values above 1 indicate a higher positive prediction rate for the group encoded as 1, which in this study corresponds to the privileged category in the default label encoding. SPD and EOD values closer to zero are interpreted as lower disparity under their respective definitions.

5.2 Main results table

Dataset	Model	Accuracy	SPD	DI	EOD
Adult Dataset	Logistic Regression	0.8057	0.1384	3.4769	0.2079
Adult Dataset	Random Forest	0.8541	0.1704	2.7552	0.0133
Adult Dataset	Gradient Boosting	0.8589	0.1653	3.0984	0.0840
Merged HR Dataset	Logistic Regression	0.8889	0.0000	1.0000	0.0000
Merged HR Dataset	Random Forest	0.7222	0.0000	1.0000	0.0000
Merged HR Dataset	Gradient Boosting	0.6667	-0.0833	0.9167	-0.1111

Table 5.1: Accuracy and fairness metrics for each model across the Adult and merged HR datasets.

5.3 Adult dataset results

Across the Adult dataset, predictive performance was strongest for Gradient Boosting, which achieved an accuracy of 0.8589. Random Forest followed closely with an accuracy of 0.8541, while Logistic Regression reached 0.8057. When fairness is considered, the pattern becomes more complex. SPD values remained positive for all three models, ranging from 0.1384 to 0.1704, which indicates that the rate of positive predictions differed consistently across the encoded gender groups. The smallest absolute SPD was produced by Logistic Regression (0.1384), but that does not mean the model was fair in every other respect. DI values were also far from parity, ranging from 2.7552 to 3.4769. Under the implementation used here, these values indicate that the group encoded as privileged received more positive predictions than the comparison group. EOD provides a slightly different perspective. The value closest to zero was obtained by Random Forest (0.0133), suggesting a narrower gap in true positive rates than the other Adult models. Taken together, the Adult results illustrate a classic fairness problem: strong predictive performance coexists with substantial group-level disparity, and the model that performs best overall does not eliminate those disparities.

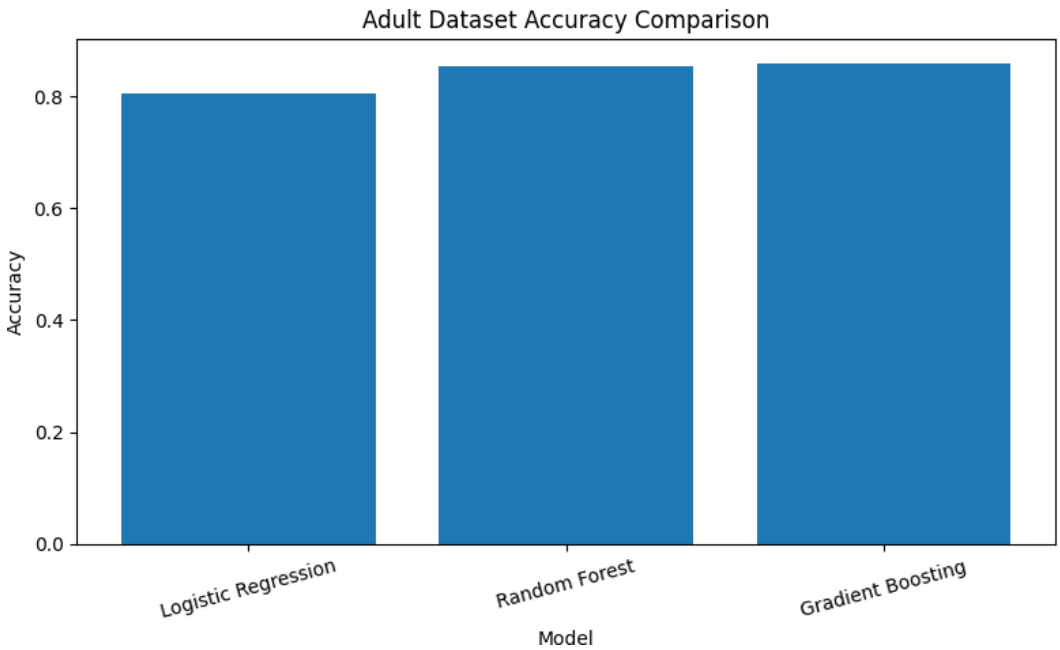

Figure 5.1: Adult dataset accuracy comparison.

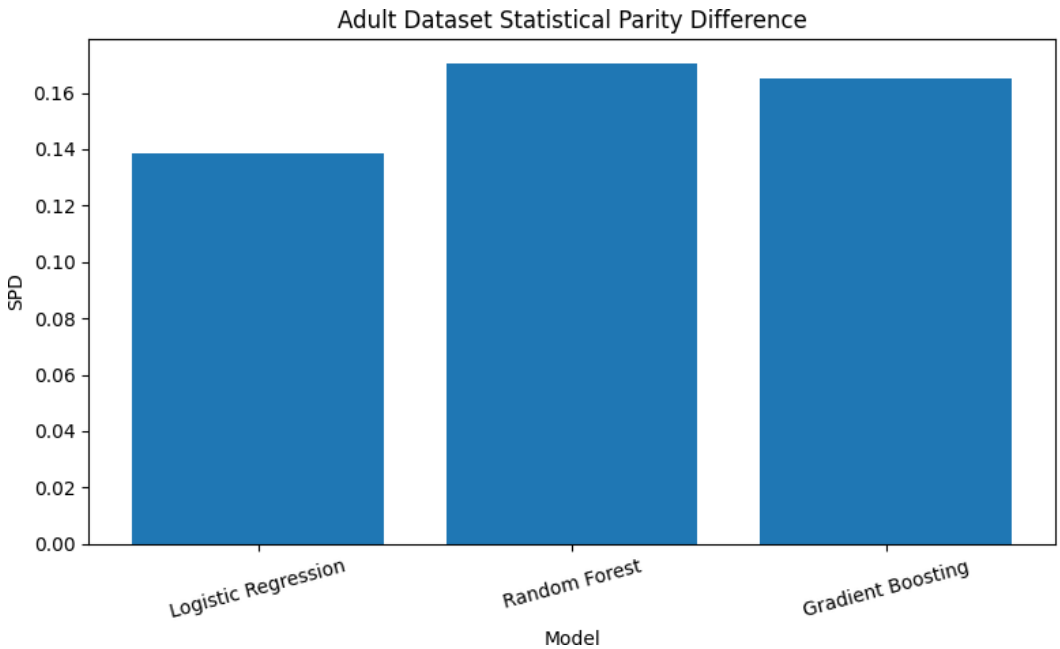

Figure 5.2: Adult dataset Statistical Parity Difference (SPD) by model.

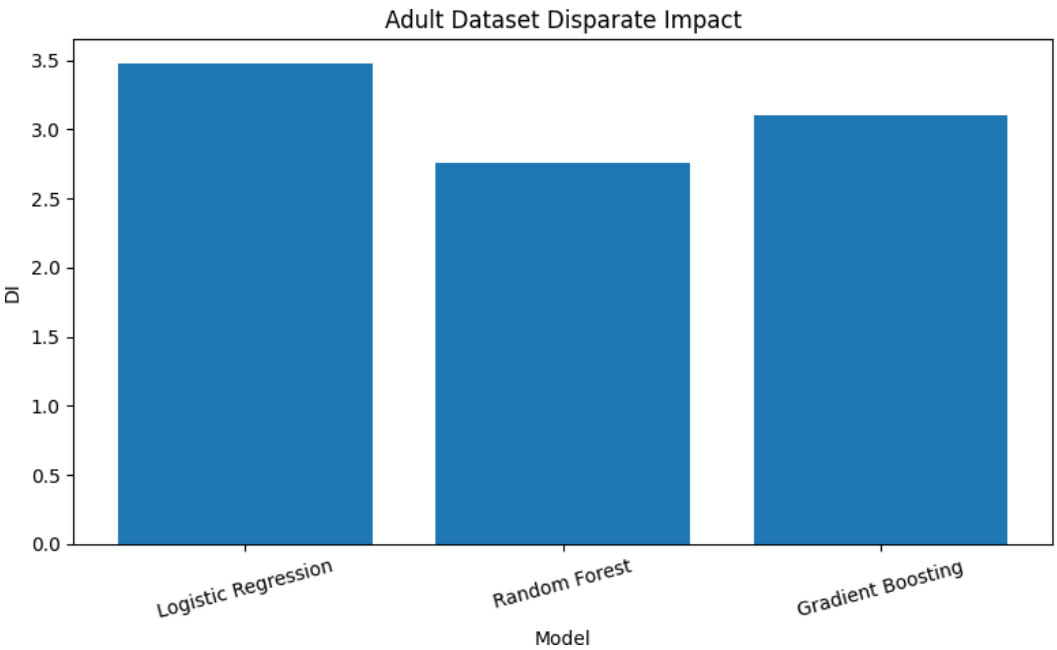

Figure 5.3: Adult dataset Disparate Impact (DI) by model.

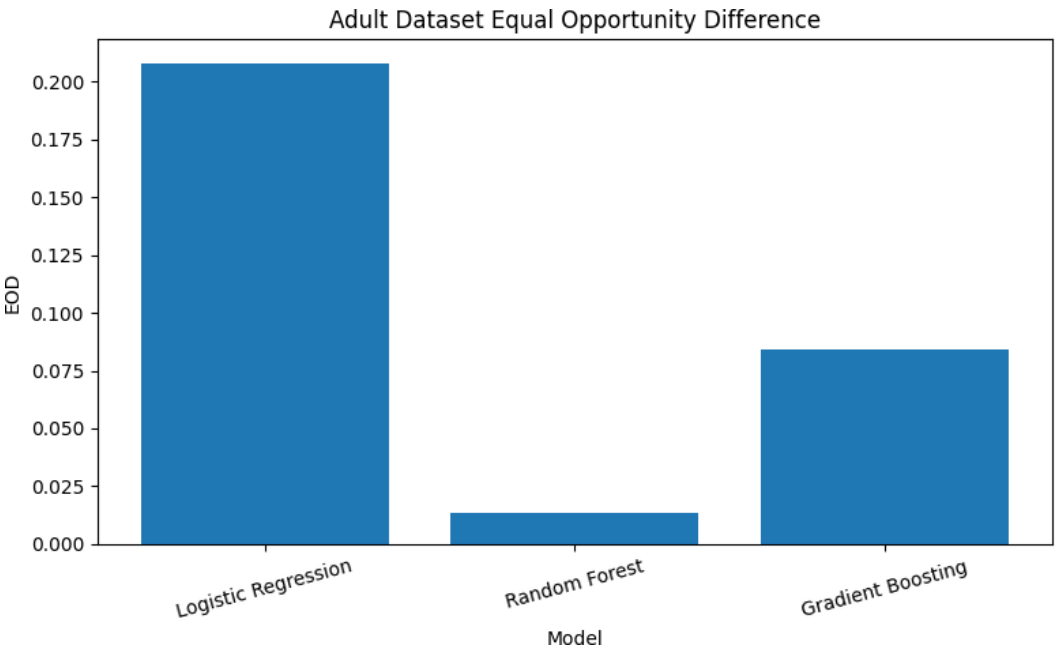

Figure 5.4: Adult dataset Equal Opportunity Difference (EOD) by model.

5.4 Merged HR dataset results

The merged HR dataset produced a different pattern. Logistic Regression achieved the highest accuracy at 0.8889, outperforming Random Forest (0.7222) and Gradient Boosting (0.6667). In fairness terms, both Logistic Regression and Random Forest returned SPD values of 0.0000, DI values of 1.0000, and EOD values of 0.0000. On the surface, these results suggest parity across the encoded gender groups. However, that interpretation must be treated carefully because the HR dataset is very small, the target is engineered from termination status, and subgroup counts are limited. Gradient Boosting produced the least favourable fairness profile on this dataset, with SPD of -0.0833, DI of 0.9167, and EOD of -0.1111, indicating that the balance of outcomes and true positive rates shifted away from parity. The contrast between the Adult and HR datasets is instructive. In the benchmark setting, disparities are clearer and more stable. In the organisational HR setting, fairness values are more fragile and therefore require more cautious interpretation.

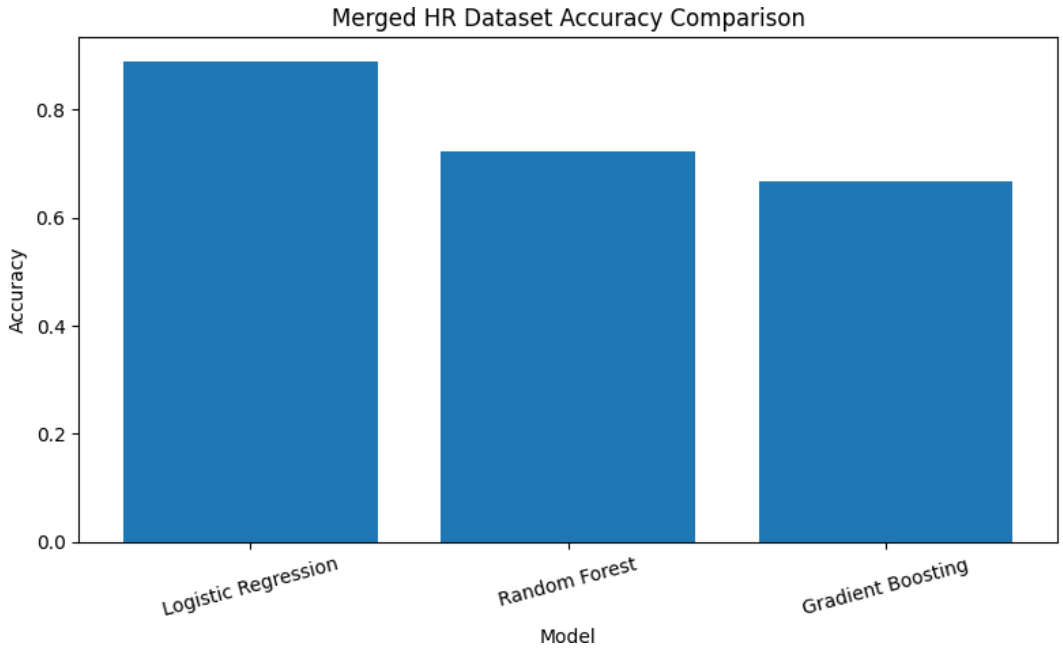

Figure 5.5: Merged HR dataset accuracy comparison.

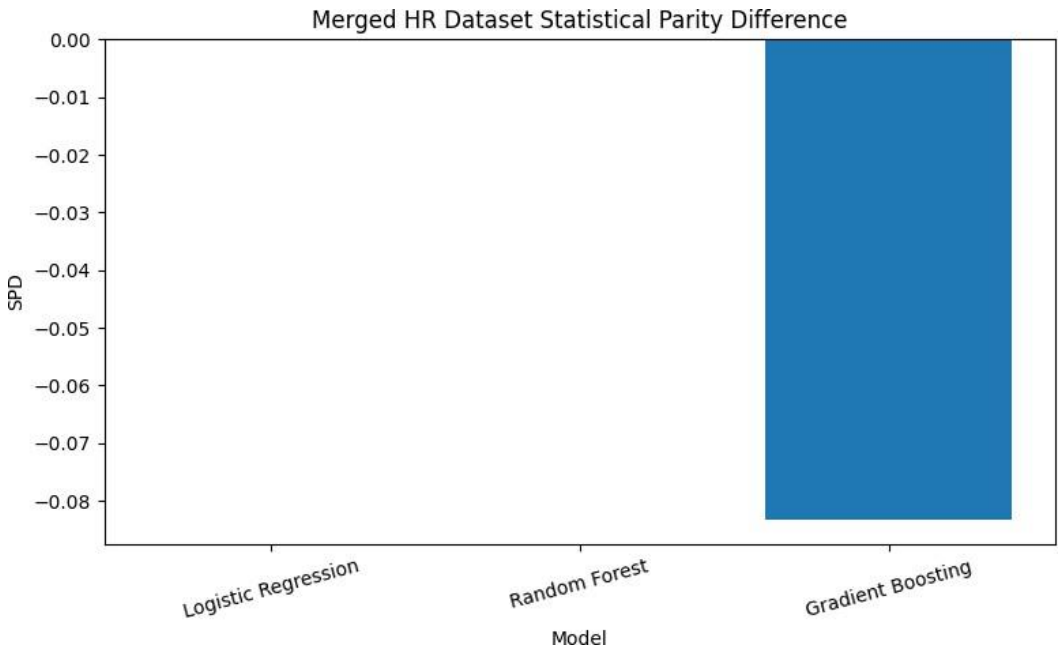

Figure 5.6: Merged HR dataset Statistical Parity Difference (SPD) by model.

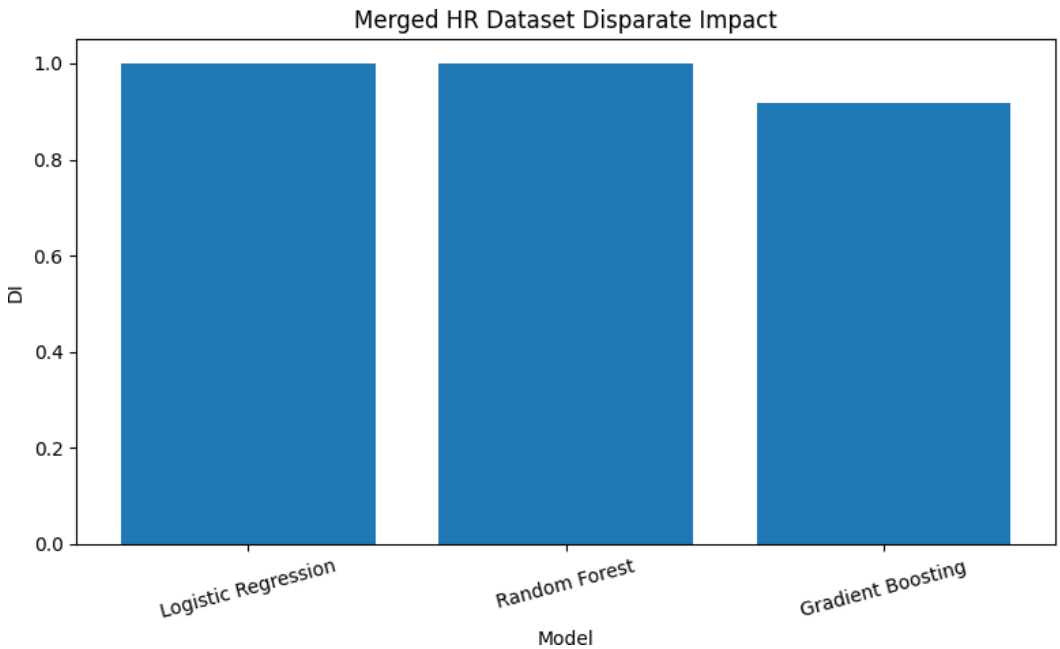

Figure 5.7: Merged HR dataset Disparate Impact (DI) by model.

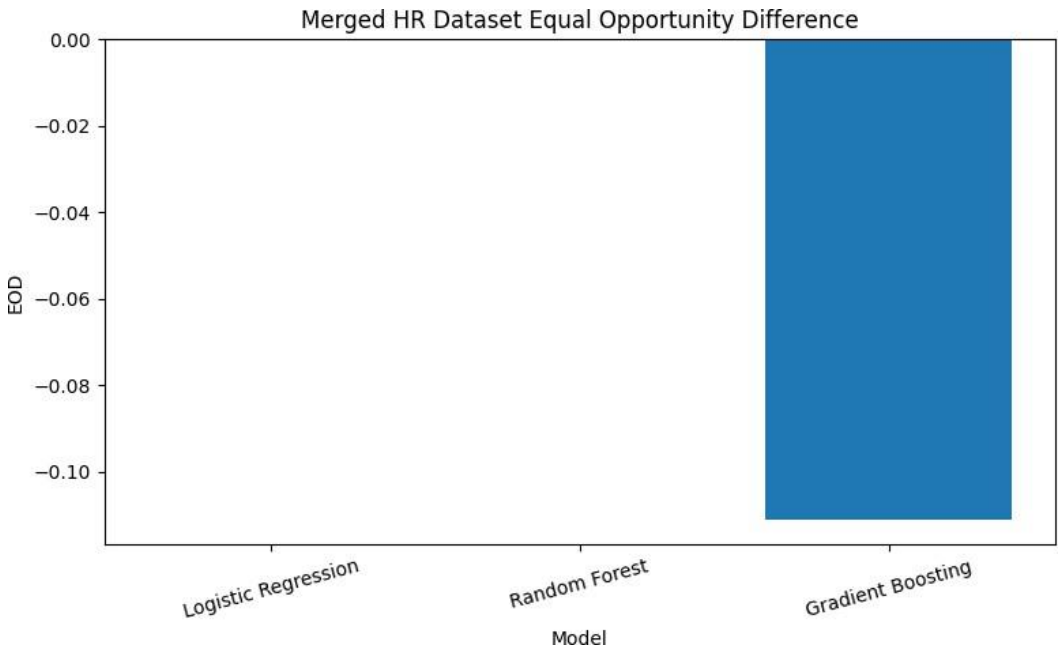

Figure 5.8: Merged HR dataset Equal Opportunity Difference (EOD) by model.

5.5 Cross-dataset synthesis

Read together, the two datasets support three conclusions. First, the benchmark Adult dataset shows that acceptable overall accuracy does not imply equitable group outcomes. Second, the HR dataset shows why fairness auditing in small organisational datasets must be handled cautiously: apparently strong fairness values may still sit on unstable subgroup counts and proxy labels. Third, the comparison demonstrates the value of a dual-dataset design. The Adult dataset provides a recognisable fairness benchmark, while the HR dataset tests how the same audit logic behaves in a more applied but more fragile setting. These findings set up the Discussion chapter, where the results are interpreted in relation to fairness–accuracy trade-offs, proxy discrimination, label validity, and governance implications.

6. Discussion

6.1 Interpreting the results

The purpose of this chapter is to interpret the fairness-audit results rather than simply restate them. Accuracy, SPD, DI, and EOD each reveal something different about model behaviour, and

none of them should be read in isolation. A model that performs well overall may still distribute positive outcomes unevenly across groups. Likewise, a model that appears relatively balanced on one fairness measure may still create disparities under another. For that reason, interpretation has to proceed through comparison rather than through a search for one definitive fairness score.

A useful starting point is to read accuracy together with fairness metrics. Accuracy tells us how often the model predicts the target correctly across the full test set. SPD and DI tell us whether one group receives favourable outcomes at a higher rate than another. EOD tells us whether positive cases are recognised equally across groups. Together, these measures reveal whether the model is simply effective in aggregate or whether it also behaves more equitably across protected groups.

6.2 Why trade-offs matter

The literature on fairness repeatedly shows that predictive performance and fairness do not always move in the same direction. A more accurate model may become better at exploiting patterns in the data, including patterns that reflect proxy relationships tied to protected characteristics. In structured employment data, variables such as department, education, salary, or tenure can carry social meaning even if they appear operationally neutral. A more flexible model can improve overall accuracy, but it may also become better at picking up patterns in the data that reflect existing group differences, which can make disparities more noticeable in its predictions (Chouldechova, 2017; Kleinberg, Mullainathan and Raghavan, 2016). At the same time, it would be too simplistic to argue that more complex models are always less fair. In some situations, a model with greater expressive power may actually perform better for underrepresented groups than a simpler baseline and, as a result, reduce certain disparities. For that reason, the relationship between fairness and performance should be treated as an empirical question to be tested, rather than as something that can be assumed in advance.

The significance of fairness–accuracy trade-offs is practical as well as theoretical. If an organisation selects models using accuracy alone, it risks normalising disparities that become visible only when group-level metrics are reported. The present study addresses this by treating fairness and performance as joint evaluation criteria.

6.3 Model complexity and proxy learning

The significance of fairness–accuracy trade-offs is practical as well as theoretical. If organisations select models using accuracy alone, they risk normalising disparities that only become visible when

group-level metrics are reported. This dissertation therefore treats fairness and performance as joint evaluation criteria, rather than allowing one to stand in for the other (Chouldechova, 2017; Selbst et al., 2019).

and interaction learning. Gradient boosting goes further by building an ensemble that can capture subtle structured patterns in the data. These differences matter because proxy discrimination may become easier to learn as model flexibility increases.

If the more flexible models show wider disparities, one plausible interpretation is that they are identifying combinations of features that reconstruct protected-group differences more effectively. If they show narrower disparities, an alternative interpretation is that the simpler baseline was underfitting important structure in ways that disadvantaged one group. In either case, fairness cannot be inferred from model complexity alone. It must be evaluated through actual outputs and interpreted in relation to the data.

6.4 Reading the Adult results

The Adult dataset functions as a benchmark in this dissertation, so its results are valuable partly because they can be interpreted alongside a wider body of fairness research. The target—income above or below fifty thousand dollars—is not itself a hiring label, but it is closely linked to labour-market inequality. Features such as occupation, education, and hours worked may correlate strongly with gender and race. Therefore, disparities in Adult predictions should not come as a surprise. They may reveal how standard classifiers respond to a dataset in which structural inequality is already embedded.

In interpreting Adult fairness results, it is especially important not to confuse benchmark status with neutrality. The dataset is useful because it is standard, not because it is socially clean. In fact, its value in fairness studies comes partly from the fact that it contains the kinds of demographic and labour-market patterns that make group disparities visible.

6.5 Reading the HR results

The HR dataset offers a different kind of value. It is closer to organisational practice, but it is much smaller and more context dependent. The target is engineered from termination status, which means the models are predicting an observable organisational outcome rather than an abstract notion of employee quality. This makes interpretation more cautious. Fairness with respect to this target does not necessarily imply fairness with respect to merit or suitability.

Because the dataset is small, subgroup fairness metrics should be read carefully. A small number of predictions can materially change SPD, DI, or EOD. This does not make the analysis

meaningless, but it does mean the HR findings should be presented as illustrative rather than strongly generalisable. In one sense, this is precisely what makes the HR dataset useful: it shows the practical difficulty of conducting fairness audits on the kinds of limited structured data that organisations often have available.

6.6 Comparison with the literature

The findings of this dissertation are consistent with several themes in the literature. First, they support the argument that fairness cannot be inferred from accuracy alone. Second, they reinforce the idea that target construction matters. In the HR dataset, fairness is being evaluated relative to an engineered organisational label, not a universal ground truth of employee worth. This aligns closely with the concerns raised in the hiring literature about what models actually predict.

Third, the findings remain compatible with the literature on proxy discrimination. Even where a sensitive feature is not the sole driver of the decision process, a model can still behave in group-differentiated ways through correlated variables. This is one reason why removing explicit sensitive attributes is not enough to guarantee fair outcomes.

Finally, the dissertation supports the sociotechnical view that fairness metrics are useful but incomplete. They help identify disparities, but they do not by themselves determine whether a system is acceptable, lawful, or just. Those judgements depend on the social meaning of the target, the deployment context, and the institutional processes surrounding the model.

6.7 Practical implications

6.7.1 Recommendations for organisational practice

The results of this dissertation point toward a set of practical recommendations that extend beyond choosing the numerically best model. The first recommendation is that employment-related machine learning systems should be documented from the beginning with fairness in mind. That documentation should record the intended use of the model, the meaning of the target variable, the protected groups considered, the metrics reported, the limitations of the training data, and the schedule for re-audit. In many real settings this information is assembled only after concerns are raised. A stronger governance approach begins the documentation process before deployment rather than after it.

The second recommendation is that organisations should define in advance what happens when fairness metrics indicate a concerning disparity. Without a response process, fairness reporting can become descriptive but practically weak. A threshold for escalation does not have to be a single rigid number, and it should not be treated as a substitute for legal advice or contextual judgement, but there should be a documented rule for when a disparity triggers review by HR, legal, or governance staff. This is especially important in employment settings, where technical metrics may have reputational and compliance implications even before they produce formal complaints.

The third recommendation is that fairness evidence should be translated for multiple audiences. Technical staff may require the raw metric values, subgroup counts, model configuration, and implementation notes. HR practitioners and managers may need a plainer explanation of what the metrics mean operationally and what kinds of harms they are intended to detect. Senior decision-makers may need a concise risk summary and a recommendation. A fairness audit has limited organisational value if it can only be understood by the person who wrote the code.

A fourth recommendation is that fairness audits should not be treated as one-time certification exercises. Data distributions shift, role profiles change, departments evolve, and organisational practices may alter the meaning of the target variable over time. A model that appears acceptable on one dataset snapshot may behave differently six months later. The practical implication is that fairness should be monitored periodically and re-evaluated when the model, the target, or the input population changes materially.

Table 6.2: Practical recommendations for organisations using employment-related machine learning systems.

Recommendation	**Why it matters**	**Minimum action**
Document model purpose and label meaning	Fairness claims weaken if assumptions are hidden	Create a short model card or audit note
Define escalation triggers	Metrics need a response pathway to matter in practice	Agree a review threshold and owner
Report to multiple audiences	Technical evidence is often unusable without interpretation	Produce technical and plain-language summaries
Repeat the audit over time	Fairness is not stable by default	Re-run after data or process changes

There is also a broader policy implication to consider. If fairness audits are treated only as technical reports produced for internal use, their ability to shape organisational behaviour may be limited. For fairness evidence to have practical value, it needs to be integrated into decision protocols. That might mean requiring fairness reporting before a model is piloted, defining escalation procedures when disparities exceed an agreed threshold, or linking model review to legal and HR oversight. In that sense, the practical implication of this dissertation is not simply that organisations should calculate more metrics. It is that they should create processes through which those metrics can matter.

A second practical implication concerns communication. One reason fairness work is often misunderstood in organisations is that technical metrics are reported without explanation. A ratio, difference, or error-rate gap may be mathematically correct but still unintelligible to non-specialist stakeholders. The dissertation therefore implicitly supports a communication model in which fairness reporting includes plain-language interpretation, explanation of what each metric captures, and an account of what remains uncertain. This is particularly relevant in employment settings, where HR teams, legal staff, and senior managers may all need to engage with fairness evidence from different perspectives.

The study also points toward a more cautious understanding of model success. In many operational settings, a model is considered successful if it performs efficiently and appears to save time. Yet an employment-related model should not be judged only by throughput or convenience. If it reduces recruiter workload but entrenches group disparities, then the efficiency gain comes at a social cost. That is why this dissertation treats fairness auditing not as an optional ethical add-on but as part of what it means to evaluate whether a model is fit for use in a sensitive domain.

Finally, the discussion supports a broader cultural point: responsible AI in employment cannot depend on faith in technical neutrality. The idea that algorithms are neutral because they are data-driven has been challenged repeatedly in both research and public debate. The findings in this dissertation belong to that wider critique. They show that model behaviour needs to be checked, interpreted, and questioned, even when the pipeline appears conventional or technically competent. In other words, fairness work is not only about detecting bad systems; it is also about resisting unwarranted confidence in ordinary ones.

The practical implication of this dissertation is that fairness auditing should be treated as part of ordinary model evaluation in employment-related systems. Organisations should not rely solely on aggregate performance when assessing algorithmic tools. They should report fairness metrics, define which fairness concerns matter most in the application, and connect those concerns to legal and governance requirements.

The dissertation also suggests that organisations need to examine the meaning of their labels. A model trained on retention, performance, or promotion outcomes is only as fair as the organisational practices reflected in those outcomes. Technical evaluation therefore needs to be paired with data governance, policy review, and human oversight.

A further issue raised by the results concerns the difference between statistical fairness and organisational fairness. Even if a model were to perform well on all reported fairness metrics, that would not automatically mean that the surrounding organisational process is fair. Selection systems include job design, criteria definition, candidate sourcing, human review, and appeals or contestation mechanisms. A fairness audit at the model level is valuable because it identifies one important layer of risk, but it cannot replace broader organisational scrutiny. This point is worth stressing because algorithmic systems are sometimes evaluated too narrowly, as though model outputs alone determine justice. In practice, harm is often produced through the interaction between the model and the institution that uses it.

.The present study measures group-level disparities. Those disparities are analytically important, but they are not by themselves legal verdicts. This distinction strengthens the dissertation because it avoids overclaiming while still taking fairness concerns seriously.

Another practical implication is that fairness audits should be linked to decision thresholds and policy choices rather than treated as one-off technical exercises. An organisation may be willing to accept a modest reduction in predictive accuracy if it significantly improves the treatment of an underrepresented group, but such a decision requires explicit justification. Similarly, if a model is retained despite known disparities, the institution should be able to explain why and

how those disparities will be monitored. In other words, fairness results should inform governance rather than remain isolated in technical reports.

The study also demonstrates why small organisational datasets require epistemic humility. It is easy to overread fairness metrics when the case counts are low, particularly if a single false positive or false negative changes a ratio or difference substantially. The HR dataset therefore serves as a useful reminder that fairness auditing is not only about detecting disparity but also about understanding the confidence and limits of the evidence. In some settings, a small dataset may still justify caution even if the fairness metrics appear benign, simply because the estimates are too fragile to support strong conclusions.

Finally, the discussion reinforces a broader point from the sociotechnical literature: fairness metrics should prompt questions as much as they answer them. If a model shows disparity, the next question is not simply how to repair the metric but why the disparity appears and whether the underlying process is itself defensible. If a model appears fair, the next question is not whether fairness has been achieved once and for all, but whether the metric used was appropriate and whether the system would remain fair under continued deployment. That mindset is arguably one of the most important outcomes of the dissertation.

6.8 Limitations of the study

6.8.1 Limitations matrix

The limits of the study are not all of the same kind. Some constrain generalisability, some affect statistical stability, and others restrict the strength of normative interpretation. Presenting them in a compact matrix helps show that the dissertation's limits are specific and interpretable rather than vague disclaimers.

Table 6.1: Structured summary of the main study limitations.

Limitation	**Type**	**Effect on findings**	**How handled here**
No direct resume-text dataset	External validity	Findings apply to structured employment-adjacent tasks rather than full applicant parsing	Scope is stated explicitly
Small HR dataset	Statistical stability	Subgroup fairness	Results are interpreted

		estimates may be fragile	cautiously
Engineered HR target	Construct validity	Fairness is relative to termination status rather than merit	Target meaning is discussed directly
Single protected attribute per dataset	Analytical breadth	Intersectional harms may remain hidden	Positioned as future work
No mitigation implemented	Project scope	Study diagnoses disparity but does not test interventions	Baseline is established clearly for later comparison

Several limitations shape the discussion. The study does not use true resume text or applicant-tracking data. The HR dataset is small. The HR target is engineered rather than normatively validated. Only one protected attribute is examined per dataset. Mitigation is not implemented. And the evaluation is based on one executed split rather than a broader sensitivity analysis. These limitations constrain the strength of the conclusions, but they do not negate the value of the audit. They simply define its scope.

6.9 Future work

The most direct next step would be to introduce mitigation and compare post-mitigation fairness to the baseline reported here. Reweighing would be a sensible pre-processing extension, while equal opportunity or equalised odds post-processing would provide a natural post hoc comparison. Further work could also include threshold analysis, repeated random splits, intersectional evaluation, or feature-level interpretability.

Another important future direction lies outside the model itself. Fairness audits become more useful when organisations integrate them into governance processes: documentation, review cycles, escalation procedures, and legal oversight. Future research could therefore connect technical fairness evaluation with organisational decision-making and accountability.

6.10 Chapter summary

The discussion has argued that the value of the results lies not only in the numbers themselves, but in the interpretive framework used to read them. Accuracy and fairness need to be examined together. Metrics capture different harms. Model complexity may interact with proxy learning.

Dataset context matters. And fairness remains inseparable from the meaning of the target and the system in which the model might be used.

7. Conclusion

This dissertation examined fairness in machine learning within an employment-adjacent setting using a reproducible audit workflow built around two datasets and three baseline classifiers. The study was motivated by the broader question of how automated shortlisting-style systems should be evaluated when their outputs may affect access to employment-related opportunity. Rather than claiming to reproduce a full industrial hiring product, the dissertation focused on a structured fairness audit that remained closely aligned with the code that was actually executed.

The project has shown that fairness cannot be inferred from aggregate performance alone. Standard classifiers may perform well overall while still producing uneven outcomes across demographic groups. It has also shown that fairness is multi-dimensional. Statistical parity, disparate impact, and equality of opportunity do not capture the same concern, and they cannot be collapsed into a single universal judgement. This is why multi-metric reporting is necessary in employment-related machine learning studies.

The dual-dataset design also proved useful. The Adult dataset offered a recognised benchmark through which model fairness could be discussed in relation to an established literature. The HR dataset, although small and limited, provided an organisational case that made the project more applied and exposed the additional interpretive challenges of real workplace data. In particular, it highlighted the importance of target meaning. Predicting retention or termination is not the same as predicting merit, and fairness results have to be read with that distinction in mind.

The dissertation has not solved fairness in hiring. It has not implemented mitigation, model governance in practice, or real resume parsing. What it has done is provide a transparent

foundation for those next steps. The results generated by the study can serve as a baseline for later interventions such as reweighing or equalised odds, and the analytical framework developed here can support more advanced work on intersectional fairness, threshold tuning, or interpretability.

The broader lesson is simple but important: fairness in machine learning should be examined deliberately, not assumed. A model that performs well overall should not be treated as acceptable without asking how that performance is distributed across groups. In employment contexts, where the consequences of classification are particularly significant, that question is not optional. It is part of what responsible machine learning requires (Selbst et al., 2019; UNESCO, 2021; OECD, 2019).

A final strength of the dissertation lies in its disciplined treatment of scope. In applied AI projects there is often a temptation to make claims that extend beyond what the data and implementation can reasonably support. This study has taken a more careful route. It has tied its arguments closely to the executed code, the two available datasets, and the fairness metrics that were actually computed. That makes the study more modest than a project claiming to solve algorithmic hiring outright, but it also makes the conclusions more trustworthy. In a field where overclaiming is common, methodological restraint is a genuine strength rather than a weakness.

The dissertation also shows that baseline work has real scholarly and practical value. Not every useful project needs to introduce a novel algorithm or a complex mitigation framework. There is significant value in demonstrating how ordinary, widely used classifiers behave when fairness is taken seriously as part of evaluation. For many organisations, even that level of auditing would represent a substantial improvement over current practice. The contribution of this study therefore lies not only in what it says about the specific datasets used, but in the research habit it promotes: evaluate standard systems critically before treating them as operationally acceptable.

One of the most useful outcomes of this dissertation is that it shows how much can already be learned from a relatively simple audit design. The project does not rely on advanced optimisation, deep learning, or a large deployment environment, yet it still produces results that are meaningful enough to support critical discussion about fairness, model choice, and dataset context. This is important because it shows that fairness evaluation should not be postponed until an organisation has built a highly sophisticated system. Even baseline models trained on structured data can display measurable disparities, and those disparities are worth examining

early. The dissertation also highlights that reproducibility has value beyond technical neatness. When preprocessing decisions, target construction, and fairness calculations are clearly documented, the resulting analysis becomes easier to interpret, question, and improve. In that sense, the study supports a broader view of responsible machine learning: not as a single fairness score or one mitigation technique, but as a process of making assumptions visible and open to scrutiny.

More broadly, the study points toward a useful middle ground in fairness research and education. Discussions of fairness often become trapped between high-level ethical statements and highly formal mathematical treatments. What students and practitioners frequently need is work that sits between those poles: technically concrete, conceptually careful, and honest about uncertainty. This dissertation aims to contribute to that middle ground. It does not reduce fairness to a slogan, but neither does it present fairness as a purely abstract optimisation problem detached from institutions. That balance is especially important in employment-related AI, where the practical

consequences of modelling choices are significant and where technical clarity must coexist with normative judgement.

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Appendices

Appendix A: Source code access

The full Python script used in this dissertation is available in the accompanying GitHub repository: https://github.com/SeverusSnapee/M599_THESIS

The executed file is `src/main.py`. The repository also contains the saved results, generated figures, dependency list, and reproduction instructions.

Appendix B: Data dictionary

HR Core (HRDatabase.xlsx) columns: id, employee, gender, birth_date, hire_date, termination_date, termination_reason, education, position, department. EmployeeInformation.xlsx columns: id, performance_review, city, last_promotion_date, salary, overdue_vacation?. DepartmentInformation.xlsx columns: department, manager. Merged HR dataset columns (pre-cleanup): id, employee, gender, birth_date, hire_date, termination_date, termination_reason, education, position, department, performance_review, city, last_promotion_date, salary, overdue_vacation?, manager. Cleaned modelling columns (post-cleanup): gender, education, position, department, performance_review, city, salary, overdue_vacation?, manager, target.

Appendix C: Reproduction instructions

Environment example: Python 3.x with pandas, numpy, scikit-learn, matplotlib, openpyxl. Install dependencies with pip install pandas numpy scikit-learn matplotlib openpyxl. Run python main t.py. Expected outputs include baseline_results_with_fairness.csv and saved plots for both datasets.